

**Time:** 5 minutes**Outcome:** Measure Profit, Loss & Break-Even.**Difficulty:** ●●○ Intermediate

Profit, Loss & Break-Even



THE CORE IDEA

Economic profit or loss equals the gap between price and average total cost multiplied by output. Profit, Loss & Break-Even measures per-unit and total economic profit or loss and interpret break-even and zero economic profit. The firm combines its revenue from price and quantity with all economic costs to measure profit. Profit/loss measurement and the short-run shutdown decision use different comparisons. Separate output, operating status, profit or loss, and long-run entry or exit pressure.



HOW TO RECOGNIZE IT

- First find the firm's chosen output using the marginal-revenue and marginal-cost rule.
- At that output, compare price with average total cost.
- Price above average total cost means profit; equality means break-even; price below average total cost means loss.
- Measure profit or loss as the per-unit gap multiplied by output.

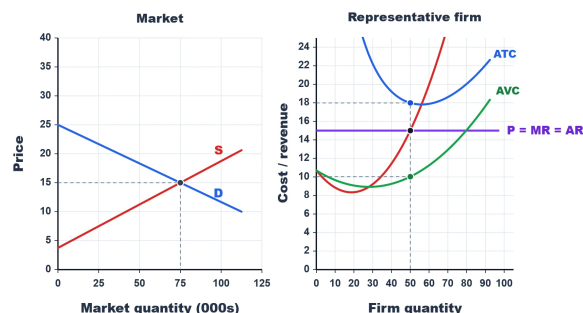


WATCH OUT

Do not confuse loss with shutdown. Profit/loss measurement and the short-run shutdown decision use different comparisons. Economic cost includes implicit opportunity costs, including the normal return needed to keep owner resources in the firm.



WORKED EXAMPLE: PROFIT, LOSS, OR BREAK-EVEN?



The market sets a price of \$15. The representative firm produces 50 units where $P = MR = MC$, but its ATC is \$18. Loss per unit is $\$18 - \$15 = \$3$, so total loss is $\$3 \times 50 = \150 . The firm may still operate because price exceeds AVC.



CHECK YOURSELF

How does the price-versus-ATC comparison connect a cost curve to the firm's profit status?

**READY?**

Return to the game and master this concept.